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3-min All-out Exercise Test for Running

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Key words

- anaerobic capacity
- critical power
- critical velocity
- maximal steady state
- $\text{VO}_{2\text{max}}$

Abstract

A 3-min all-out exercise test (3 MT) estimates critical power and the curvature constant for cycle ergometry validly; however, the mode of running has not been studied. We examined the efficacy of a running 3 MT, using global positioning sensor data, to predict outdoor racing performance. Women distance runners ($n=14$) were tested at preseason within a month prior to competing officially in either short or middle distance races. Critical speed (CS) (4.46 ± 0.41 m/s) estimated from the 3 MT did not differ ($p > 0.05$) from the mean speed of gas exchange threshold

and maximum oxygen uptake ($50\%\Delta$), as derived from a custom treadmill graded exercise test (4.55 ± 0.24 m/s). Runners with higher curvature constants (D'), estimated from the 3 MT, raced at higher speeds above CS (R^2 ranging 0.63–0.99). Race speeds for 800 m exceeded the speed for 150 s of all-out running, rendering 800 m estimates less accurate. Conversely, predicted times for the other distances yielded strong intraclass correlations (α) and low coefficients of variation (%) values ($\alpha=0.74/1.7\%$ and $\alpha=0.87/2.1\%$, for 1 600 and 5 000 m, respectively). Use of the running 3 MT for performances ranging ~2.5–18 min is recommended.

Introduction

Short-term, exercise performance ranging ~2–15 min is characterized as the severe exercise domain [31]. The lower limit of the exercise intensity for the severe domain is defined by the maximal steady state (MSS) for lactate and oxygen uptake (VO_2) and the upper limit is defined by an intensity too extreme to allow sufficient time for VO_2 kinetics to achieve maximum ($\text{VO}_{2\text{max}}$) [22]. The mechanical metric associated with the MSS is critical power (CP) or critical speed (CS), for cycling and running, respectively [6]. CP is the approximate mean value ($50\%\Delta$) for power evoking gas exchange threshold (GET) and $\text{VO}_{2\text{max}}$, as determined from a graded exercise test (GXT) [5]. That is, when exercising intensity surpasses CP or CS, aerobic pathways can no longer sustain muscle ATP demands exclusively [6]. Indeed in the severe domain, expenditure of a finite anaerobic energy reservoir occurs concurrently with a rise in VO_2 kinetics toward $\text{VO}_{2\text{max}}$ [22]. Where $\text{VO}_{2\text{max}}$ identifies a limit of aerobic power, GET as a percentage of $\text{VO}_{2\text{max}}$, has been reported as a more sensitive predictor of endurance performance [8]. For running, performance in the

severe domain relies on both CS and anaerobic capacity [28,29]. In cycling, the anaerobic capacity is defined by maximal capacity for work above CP (W'). In running, the anaerobic capacity is defined by maximal capacity to displace the body (D') at speeds above CS. For instance if one's CS is 4.0 m/s and D' is 120 m, that person's ability to sustain a running speed of 5 m/s (1 m/s faster) would terminate at 120 s (i.e., 120 m/1 m/s above CS = 120 s to deplete D') [22]. Thus, in theory, the superior method for predicting performance in the severe domain is to estimate CS and D' . Power/running speed and time to fatigue has been described as a hyperbolic relationship [14,24,29]. Indeed, work in the severe domain conforms to Newton's work-energy theorem such that as one increases work above CS, one increases depletion of D' proportionately. Similarly, the rise of VO_2 and time to $\text{VO}_{2\text{max}}$ is proportionately more rapid at higher intensities above CS, and has been described similarly as a hyperbolic relationship [16]. The traditional method of deriving CP and W' involved examining a series of times to exhaustion at various intensities relative to peak power [9]. Early debate centered on the number of bouts that

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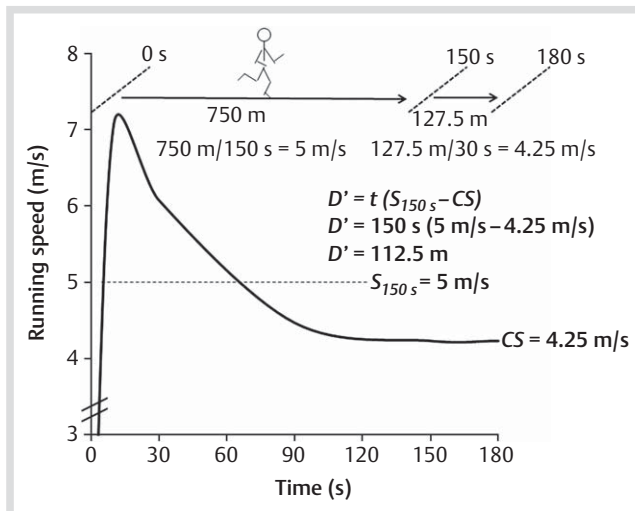


Fig. 1 GPS tracking of a 3-min all-out running test for a representative subject.

should be performed for valid modeling [15]. More recently, a concern for the high variability of time to fatigue at a given intensity has been expressed [38], with such error being compounded in the estimate of W' . Conversely, a 3-min all-out exercise test (3MT) has been developed which estimates the MSS, or CP [5], and W' [34] for cycling. The 3MT yields reliable measures [19] and can replace the more labor-intensive traditional method of gathering data on multiple, constant-load bouts. With an ability to measure displacement relative to time using global positioning sensor (GPS) data, it is conceivable to conduct a running 3MT to derive CS and D' ; however, such a technique has not been reported. Therefore, we examined the efficacy of a running 3MT with GPS on a group of competitive distance runners.

Materials and Methods

Subjects

14 collegiate women distance runners were tested at pre-season (age: 19 ± 1 year; bioelectric impedance: $15 \pm 3\%$ body fat; VO_{2max} : 55 ± 4 ml/kg/min). All subjects completed a waiver acknowledging compliance with the USA National Collegiate Athletic Association guidelines, a waiver in which athletes swear to abstain from using a list of banned substances. Subjects also verified they avoided running for 24 h preceding each test in this study. As part of their competitive status, all subjects were deemed healthy from a recent physical examination performed by a physician. All tests associated with this study were scheduled as part of workouts occurring regularly to minimize disruption of their training regime. Subjects provided informed, written consent to participate. All procedures were pre-approved from our institutional review board, in accordance with ethical standards of the International Journal of Sports Medicine [12].

Theoretical model

The 3MT presumes that a runner will expend D' within 2.5 min of all-out effort whereby the mean speed between 2.5 and 3.0 min will reach a nadir at CS [5, 34]. Thus, D' from a running 3MT is derived using:

$$D' = t(S_{150s} - CS) \quad [a]$$

Where time (t) equals 150 s, S_{150s} (m/s) equals the average speed for the first 150 s, CS (m/s) is the average speed between 150 s and 180 s (● Fig. 1). As such, vertical and horizontal axes of the running speed-time (S - t) relationship for the severe domain are modeled using:

$$S = (D'/t) + CS \quad [b]$$

and

$$t = 1/[S - CS]/D' \quad [c]$$

or

$$t = (D - D')/CS, \quad [d]$$

where t is expressed in s, speed (S) and CS are expressed in m/s, and distance (D) and D' are expressed in m [37].

3-min all-out exercise test

The 3MT was conducted on a level outdoor track with minimal wind conditions and a clear sky. Subjects were instructed to build up to the maximal speed and maintain as fast of a running speed throughout the entire test.

We instructed subjects that the test would last approximately 3 min. Verbal encouragement was provided; however, subjects were not informed of the time-elapsed or time-remaining to discourage pacing. Subjects were told the test was over when 3 min and 5 s elapsed to ensure gathering a full 3 min of GPS data.

Each subject wore a GPS wrist watch (Forerunner Model 305, Garmin, Taiwan) and the company's commercial software was used to derive displacements covered at 150 and 180 s. Using the tcx file extension, data were exported and opened in Microsoft® Excel®. When the GPS receiver is activated, accuracy of the measurement (nearest m) appears once satellites are located and signal is received. Accuracy for the GPS receivers for the running 3MTs was ~3 m.

Graded exercise test with verification for VO_{2max}

As pacing is a concern of the 3MT, we also conducted a GXT for the purpose of identifying the speed evoking gas exchange threshold ($sGET$), the speed evoking VO_{2max} (sVO_{2max}), and 50% Δ . In the cycling 3MT, the power at 50% Δ is used to calculate the linear factor for assigning an appropriate load. The subject is anticipated to have an end-exercise cadence similar to that which is preferred when performing a GXT (~90 rpm) – i.e., a power in close proximity to 50% Δ . For the running 3MT, our belief was that 50% Δ would allow us to screen if subject's paced during the 3MT. For instance, a CS from a 3MT exceeding 50% Δ , from the GXT, would denote an inflated CS from pacing and would warrant retesting. Retesting was necessary for 3 subjects and all GXT testing was performed within 1 wk of the 3MT. A calibrated treadmill (PreCore, model C966, USA), capable of adjusting 0.5% grade and 0.045 m/s increments, was utilized for the GXT. A speed-limit of 3.98 m/s was imposed to avoid compromising the subject's running form and to prevent accidents on the treadmill belt. A rate of 0.045 m/s was subtracted from treadmill speed to approximate outdoor running; whereby, level-grade, outdoor running was presumed equivalent to treadmill running at a 1% grade [21].

Safety, equipment, and desired GXT duration (10–14 min) factored into the multiple computations we used to customize the GXT protocol for this group of athletes. First, we selected 3.98 m/s as the speed-limit, for safety purposes, a speed-limit below the team's mean CS as derived from the running 3MT. As such, a

Table 1 Custom graded exercise test protocol with outdoor running speed equivalents.

Time	Speed	Grade	Outdoor Running Speed Equivalent
(min)	m/s	%	m/s
0	2.90	0	2.88
1	3.08		3.06
2	3.26		3.23
3	3.44		3.41
4	3.62		3.59
5	3.80		3.77
6	3.98		3.95
7		1	4.13
8		2	4.31
9		3	4.48
10		4	4.66
11		5	4.84
12		6	5.02
13		7	5.20
14		8	5.38

grade computation also was necessary and we therefore assigned the imposed speed-limit to begin at 6 min, projecting treadmill grade/speed corresponding to CS for outdoor running to occur after 9 min. Third, we calculated that a +1%/min rise in grade, above 3.98 m/s, would translate to a rate increase of 0.18 m/s per 1-min stage, as described previously [32]. Thus, a starting speed of 2.90 m/s was resolved (● **Table 1**). We projected that sVO_{2max} and VO_{2max} would occur at ~12 and 13 min, respectively – 3 stages subsequent to CS (9 min), expecting the fitter athletes to require an extra grade. With these factors in mind, we expected the GXT to last between 10 and 14 min. Such a GXT range has been reported as suitable for yielding consistent bout VO_{2max} values for a single visit GXT-verification protocol [25].

A rating of perceived exertion (RPE), 6–20 range [3], was obtained prior to advancing each stage and was used to monitor when subjects felt close to exhaustion. Strong verbal encouragement was provided to ensure full effort. Subjects signaled their exhaustion with a thumb-up gesture and at which point were assisted to straddle the treadmill belt. Speed was returned to 2.9 m/s and level grade and subjects resumed running for a 3-min cool-down period. Upon completion of the cool-down period, speed was advanced to 3.98 m/s at a grade adjusted to a value of the highest grade achieved during the GXT minus a 2% grade. The intent of the subsequent fixed speed/grade, exhaustive bout was to verify VO_{2max} .

Expired breath-by-breath gas exchange data were collected on a metabolic analyzer (Parvomedics, Logan UT, USA) and exported using 15 s averaging [33]. Calibration prior to each test was performed according to the manufacturer's instructions. GET was determined using the v-slope method [2]. The $sGET$ and sVO_{2max} were interpolated linearly as 1 min preceding the sample. For instance, if the v-slope analysis determined that GET occurred at 7 min, the outdoor running speed equivalent associated with $sGET$ was interpolated as 3.95 m/s (● **Table 1**, outdoor running speed equivalent column). 50Δ for the outdoor running speed equivalent was derived as the mean of $sGET$ and sVO_{2max} .

Official race data

Official times for 800, 1 600, and 5 000 m outdoor running races were obtained; however, not each subject competed each event

Table 2 Summary statistics ($M \pm SD$) for the graded exercise, square-wave verification, and 3-min all-out running tests.

Parameter	Result
aCS	4.46 ± 0.41 m/s
D'	85.8 ± 40.5 m
$sGET$	3.93 ± 0.23 m/s
$^a50\Delta$	4.55 ± 0.24 m/s
sVO_{2max}	5.17 ± 0.24 m/s
$^bVO_{2max}$ (GXT)	54.8 ± 3.3 ml/kg/min
$^bVO_{2max}$ (Square-wave)	54.6 ± 3.7 ml/kg/min

a Not significantly different ($t = 0.94$, $p = 0.37$)

b Not significantly different ($t = 0.33$, $p = 0.75$)

(N.B., GPS was not worn for races). The 800 and 1 600 m races were outdoor tracks whereas the 5 000 m event was a cross-country race regarded as a course with only mild changes in gradations. Although Internet searches of weather indicated each race was of moderate temperature, humidity, and low wind, we did not measure direct environmental conditions for these races. The 3 MT and treadmill GXT tests were performed within a month of official race times.

Statistical analyses

Trial comparisons were made using paired t-tests. Relative consistency between tests was evaluated using intraclass correlation coefficients (ICC α) whereas absolute consistency was evaluated using coefficient of variation (CV%) and typical error [17]. D' and speed above CS for each race were evaluated with simple linear regression (R^2). Summary statistics are reported as $M \pm SD$.

Results

The group's CS, as derived from the outdoor 3 MT, corresponded to a speed between the treadmill GXT values for $sGET$ and sVO_{2max} (i.e., CS was not significantly different from 50Δ as reported in ● **Table 2**). Thus, pacing in the 3 MT was not a concern for our results. Adjustment of grade, instead of speed, at 7 min into the GXT maintained a linear increase in VO_2 relative to time (● **Fig. 2**). As such, with 15 s averaging being used, precision from the GXT was 0.045 m/s (i.e., 0.18 m/s divided by 4 samples). GXT duration was 12.0 ± 1.5 min and the verification bout duration was 2.4 ± 0.7 min. Relative VO_{2max} values between the GXT and verification bout did not differ (● **Table 2**) and were internally consistent (CV%=2.8, typical error=1.47 ml/kg/min). Thus, true VO_{2max} and subsequently sVO_{2max} were identified. The $sGET$, as derived from the GXT, was $88.6 \pm 6\%$ (~90%) of the CS estimated using the 3 MT. The sVO_{2max} value, known from the GXT, occurred at ~90 s into the 3 MT.

Estimated maximum times were compared with official race times for the following distances: 800 m ($n=5$), 1 600 m ($n=5$), and 5 000 m ($n=6$). ● **Fig. 3** displays $D-t$ and $S-t$ relationships for the estimated and official times. In comparison to 800 m, estimates from the 3 MT were more accurate for the middle distances of 1 600 m and 5 000 m, within time ranges of ~5–18 min (● **Table 3**). Moderate to very strong relationships were observed between D' and the race speed maintained above CS for each race (● **Fig. 4**).



Discussion

The primary findings of this study are as follows. A 3 MT with GPS along with the 2-component model equations (equations a through d) can estimate outdoor race performance within ~2% for 1600 and 5000m with a larger error being observed for 800m races. Those with higher D' values, relative to their peers

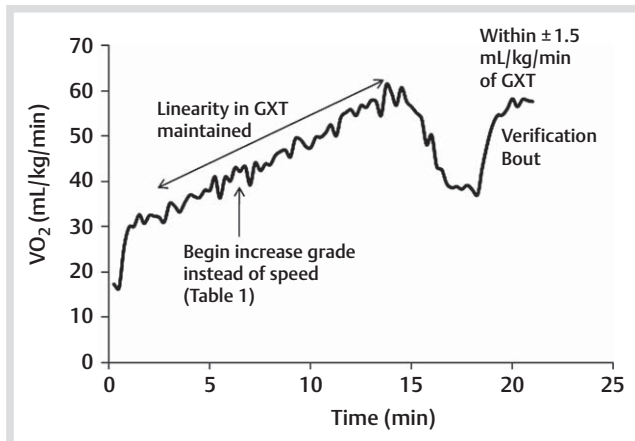


Fig. 2 Oxygen uptake relative to time for a representative subject's graded exercise test and square-wave verification. Arrow denotes when 0.18 m/s per min increase in speed was replaced with a 1% per min rise in grade.

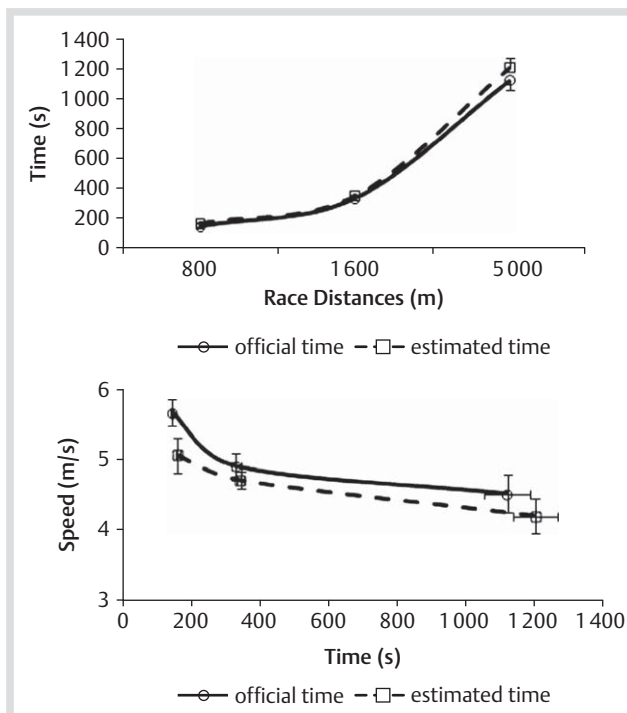


Fig. 3 Official and estimate D-t (top) and S-t (bottom) performances ($M \pm SD$).

in the same race, completed races at higher net speeds above CS, offering an added finding of proof of principle for the 3 MT.

The CS values from the running 3 MT were corroborated with our treadmill GXT value for 50% Δ (Table 2). Taking into perspective the precision of our GXT (0.045 m/s from a 15 s sampling average), the CS and 50% Δ difference (0.09 m/s) equated to a 30 s difference between incremental and constant speed running. Concerning the outdoor race time predictions, the 3 MT estimated accurate times for distances of 1600 m and 5000 m; whereas the model was less accurate for 800 m due to a too short time period (142 ± 4 s). As indicated in equation a and in Fig. 1, the estimate of D' relies on S_{150s} and therefore speeds exceeding S_{150s} should not be estimated with the 2 component model. A nonlinear model [11] with maximum running speed (S_{max}) may be used to estimate distances of 800 m or less. Prior researchers [4] have used the speed for sustaining 8 steps without backward drift on the treadmill as an index of S_{max} ; however, their metric of maximal aerobic power was derived from a series of exhaustive bouts using Douglas bags (i.e., labor-intensive). Use of data from the 3 MT along with S_{max} might yield a more time-effective means of estimating race times for distances of 800 m or less. One should keep in mind any predictive time/speed for a race time based on the 3 MT has an inherent error associated with it. In our sample, the time prediction error for running 1600 and 5000 m was ~2% (Table 3). The CP estimate, from the cycling 3 MT is consistent, on a test-by-test basis [5, 19, 34] and is uninfluenced by fatigue [35]. The parameter of W' , and presumably D' , may be less sensitive and vulnerable to day-to-day variations in glycogen storage [27]. Indeed the test-retest reliability of W' is poor; however, when modeled with CP, can still yield consistent predictive times [19]. In the present study, we were unable to conduct a reliability analysis and explore a wide range of official race times due to the subjects' practice and competitive schedule. Clearly, more research on the error associated with estimates from the 3 MT is warranted.

Another consideration for using the 2-component model from the 3 MT is to consider its upper time-limit for accuracy. A duration of 15 min has been suggested previously for sustaining CS [31]; although one group reported CS could conceivably be sustained for 30 min based on their reading of earlier research [36]. We had official 5000 m race times on 5 subjects, each of whom completed those races at velocities ~0.07 m/s above CS, in ~1100 s. The typical error for speed in the 5000 m race on these same subjects was 0.09 m/s (Table 3). Although it is conceivable that these runners could sustain speeds above CS for longer distances, the narrow speed above CS is below the typical error for our predictive model. We therefore recommend using the running 3 MT to estimate race time for distances ranging from 1000 m up to 5000 m, or an upper limit for predicting severe exercise performance at ~18 min.

As indicated in Fig. 4, runners with higher D' values, in comparison to runners with lower D' values, ran at higher speeds relative to their CS. The slope of the S-t relationship varies such that those with higher D' values (i.e., different curvature con-

Table 3 Inferential statistics of estimates derived from the 3-min all-out running test.

Distance (m)	ICC (alpha)	CV%	Typical Error Velocity (m/s)	Typical Error Time (s)	Lower 95% CI Time (s)	Upper 95% CI Time (s)
800	0.65	5.4	0.28	7.96	5.17	18.89
1600	0.74	1.7	0.08	5.59	3.63	13.27
5000	0.87	2.1	0.09	23.52	15.80	49.13

ICC – intraclass correlation; CV% – coefficient of variation. Note: descriptive statistics are depicted in Fig. 3

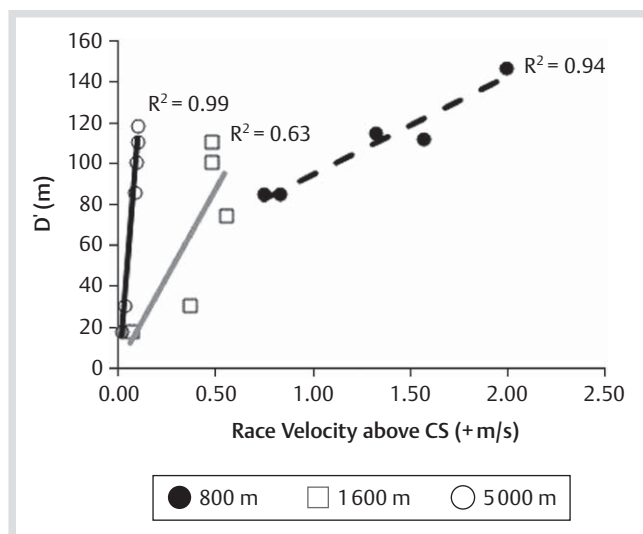


Fig. 4 Relationship between D' and the race speed above critical speed (CS) as estimated from the 3 MT. Those with higher D' values utilized a strategy of racing at a higher speed above CS.

stants) are capable of running at higher speeds above CS before fatigue ensues. Despite running at higher speeds above CS, the rate at which VO_2 approached maximum was presumably similar to those running at lower speeds above CS, because the expenditure of D' is coupled tightly with VO_2 kinetics [16] (for review, see ref. [6]). Less is known about how the runners with higher D' values in this study were able to self-select higher race speeds relative to CS (note: we did not inform the runners of their projected times). A plausible explanation may be how the runner's perceive muscular acidosis. For example, Jones et al. [23] demonstrated recently that the expenditure of W' corresponded with a time-dependent decline in muscular pH. Subjects in their study all exercised at 10% above CP; yet, the variability of W' between subjects resulted in a wide range of exhaustion times (i.e., 6.9–23.8 min). Data on perception of effort and the time-dependent rise in VO_2 (toward maximum) comes from the results in Herman et al. [13]. In their study, subjects' RPE increased with the time-dependent rise in VO_2 (i.e., the VO_2 slow component). Moreover, their subjects also were able to self-select a decline in power, in time-dependent fashion, to attenuate the VO_2 slow component, as evidenced by a steady-state of RPE. Thus, runners with higher D' values may run intuitively at higher speeds, relative to CS, based on chemoreceptive feedback from the exercising muscles [1]. Regardless of the mechanism, the data in **Fig. 4** is one more piece of evidence supporting the validity of the running 3 MT.

With longer running distances, specifically those longer than 5000 m, the metric of D' adds little predictive value. Conversely, the metric of CS corresponds with $s\text{GET}$ and is a metric more predictive of longer distances [8]. In a study on trained men, Francis et al. [10] reported that the cycling GET occurred at ~75% of CP, based on a 3 MT for cycle ergometry. As both their study and our study used trained subjects, the disparity between our findings is most likely related to differences in the mechanical efficiency between running and cycling. Indeed, running $s\text{GET}$ and CS have been described to occur at higher percentages of $\text{VO}_{2\text{max}}$, in comparison to cycling, because a greater portion of the running cycle utilizes eccentric contractions [20]. We recommend that if only running 3 MT data are available, $s\text{GET}$ may

be estimated at 90% of CS and used to project race times for distances longer than 5000 m. Of course, empirical data on the error associated with time estimates of longer distances is needed.

To attain valid measures of $\text{VO}_{2\text{max}}$, a GXT with a verification bout is preferred [26]. To our knowledge, we are the first to report a verification protocol on trained athletes. In designing the protocol, we aimed to derive a GXT that was safe (i.e., a treadmill speed above $3.98 \text{ m}\cdot\text{s}^{-1}$ was impractical for these subjects), of sufficient duration to allow both GET and $s\text{GET}$ detection, and allow for an adequate duration of the verification bout (~2 min). Robergs [32] described the basis of deriving custom GXTs; however, his model presumed an estimate of $\text{VO}_{2\text{max}}$ is known. Where non-exercising estimates of $\text{VO}_{2\text{max}}$ exist [18], such equations for athletes are unavailable. In the present study, we utilized the mean CS results of the 3 MT to design our GXT protocol. Consistency between the GXT and verification (i.e., less than 1.5 ml/kg/min) were comparable to data reported on recreational men [25]. We also observed that $s\text{VO}_{2\text{max}}$ occurred at ~90 s into the 3 MT (**Fig. 3**). Thus, the 90 s speed value in the 3 MT may serve as a useful predictor of $s\text{VO}_{2\text{max}}$ for future investigators desiring to derive a custom GXT protocol. For instance, if a $s\text{VO}_{2\text{max}}$ was estimated at 4.84 m/s , and a 10 min protocol was desired, the protocol would begin at a treadmill speed of 3.08 m/s (**Table 1**). The custom GXT along with a running 3 MT is a useful pairing of tests because $50\%\Delta$ may be used to determine if CS is potentially inflated. Our procedures were to retest the 3 MT if the CS estimate surpassed the outdoor running speed equivalent for $50\%\Delta$.

Although 800 m was a distance too short for predicting race times, the 2-component model can be used to conceivably prescribe 800 m interval training times for women distance runners (note: most likely 1000 m intervals for men) [7]. Specifically, equations b and c can be manipulated for interval training [30] using:

$$\text{Interval } t = (D - \text{Interval } \% \cdot D') / \text{CS}, \quad [e]$$

where Interval % is expressed as a fractional percent, for instance, 80% depletion of D' would be $D' \cdot 0.80$. Using our sample's mean results for CS and D' , 4.5 m/s and 85 m , the target for an 800 m interval depleting 0.80 of D' would be 163 s (4.92 m/s), or a time exceeding 150 s (Note: 80% for this example was selected arbitrarily). The accuracy of the 2-component model (equations b and c), using 3 MT data, is therefore preserved because the interval is based on a projection of depleting D' completely across a longer distance – the distance in our example being 1000 m, in accordance with the kinetic energy equation. In other words, if we presume that the running time for a 1000 m race can be estimated validly using the 3 MT, because time for that longer distance exceeds 150 s , the interval derived for a shorter proportional distance also is valid (i.e., 1000 m at 4.92 m/s , in our example, would deplete D' completely, as corroborated by solving the t of 203.3 s , with equation d, where $0.8 \cdot 203.3 \text{ s} = 163 \text{ s}$ or $0.8 \cdot 1000 = 800 \text{ m}$).

In the present study, we used GPS to determine the parameters of CS and D' . Resultant data from GPS are in displacement (vector) and not distance (scalar) units. Thus, estimates from GPS measures are velocity; however, we have used the term speed for the sake of consistency. The predictions from the 3 MT, therefore, presume that race displacement and distance is equivalent, which is seldom the case (i.e., races are often won by the runner



with the lowest net displacement) [6]. The technology of GPS is limited by sampling rate, which was typically every 2–5 s. Thus, if one simply wanted to account for the error of the GPS signal for a 1 000 m race, an error of 3 m would be added and subtracted from the displacement used in the 2-component estimates (i.e., a range of 997–1 003 m). The common alternative devices for measuring displacement are less effective which was why we selected GPS. Pedometers, for instance, register steps without consideration of change in step length. Accelerometers are problematic for measuring displacement when speed is constant; which is the case when CS is established at the end of the 3 MT. Video represents a possible option; however, the labor and equipment of digitizing adds time and cost to the test. Thus, at present, GPS represents the most suitable technology for conducting outdoor running 3 MTs. With multiple GPS watch receivers, testing a group of 20 athletes can be done in under an hour.

In summary, the 3 MT appears equally effective for running, in comparison to cycling, and does not require a preliminary GXT before its implementation (i.e., no need for a 'linear factor' calculation). Data from the 3 MT may be used in establishing custom GXTs and the combined data from the 3 MT and GXT can provide a comprehensive analysis of a runner's physiological status. The 3 MT also represents a convenient means of prescribing interval training (equation e). Future research should explore the utility of the 3 MT with other populations and with emerging technologies.

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